

Chapter-4 Curve Sketching

• Even Function

↳ A function $f: A \rightarrow B$ is said to be even if $f(-x) = f(x)$ for all $x \in A$.

e.g: 1. a) $f(x) = 2x^2 + \sin^2 x + 1$
 $f(-x) = 2(-x)^2 + \sin^2(-x) + 1$
 $= 2x^2 + \sin^2 x + 1$

or, $f(-x) = f(x)$

$\therefore f(x)$ is even function.

• Odd Function

↳ A function $f: A \rightarrow B$ is said to be odd if $f(-x) = -f(x)$ for all $x \in A$

e.g: f) $f(x) = 3x^3 + 5x$
 $= 3(-x)^3 + 5(-x)$
 $= -3x^3 - 5x$
 $= -(3x^3 + 5x)$

or, $f(-x) = -f(x)$

$\therefore f(x)$ is odd function

• Symmetry

i) Symmetry about y-axis

- A curve represented by $y = f(x)$ is said to be symmetric about y-axis if it is even function.

i.e. $f(x) = f(-x)$

Exercise 4.1

2. Examine whether the following functions are even or odd or neither.

a) $f(x) = 2x^2 + \sin x + 1$

$$f(-x) = 2(-x)^2 + \sin^2(-x) + 1$$

$$= 2x^2 + \sin^2 x + 1$$

or, $f(-x) = f(x)$

$\therefore f(x)$ is even function.

b) $f(x) = x^3 + \cos x + 1$

$$f(-x) = (-x)^3 + \cos(-x) + 1$$

$$= -x^3 + \cos x + 1$$

$\therefore f(x)$ is neither even function nor odd function.

c) $f(x) = x^2 - 4x + 7$

$$f(-x) = (-x)^2 - 4(-x) + 7$$

$$= x^2 + 4x + 7$$

$\therefore f(x)$ is neither even nor odd function.

d) $f(x) = \sqrt{x^2 + 1} - \sqrt{x^2 - 1}$

$$f(-x) = \sqrt{(-x)^2 + 1} - \sqrt{(-x)^2 - 1}$$

$$= \sqrt{x^2 + 1} - \sqrt{x^2 - 1}$$

or, $f(-x) = f(x)$

$\therefore f(x)$ is even function.

e) $f(x) = x \sin x + \cos x$

or, $f(-x) = -x \sin(-x) + \cos(-x)$
 $= x \sin x + \cos x$

or, $f(-x) = f(x)$

$\therefore f(x)$ is even function.

f) $f(x) = 3x^3 + 5x$

$f(-x) = 3(-x)^3 + 5(-x)$
 $= -3x^3 - 5x$
 $= -(3x^3 + 5x)$

or, $f(-x) = -f(x)$

$\therefore f(x)$ is odd function.

g) $f(x) = \frac{x}{x^2+1}$

$f(-x) = \frac{-x}{(-x)^2+1} = \frac{-x}{x^2+1}$
 $= -\left(\frac{x}{x^2+1}\right)$

or, $f(-x) = -f(x)$

$\therefore f(x)$ is odd function.

h) $f(x) = e^x + e^{-x}$

$f(-x) = e^{-x} + e^{-(-x)}$
 $= e^{-x} + e^x$
 $= e^x + e^{-x}$

$f(-x) = f(x)$

$\therefore f(x)$ is even function.

ii) Symmetric about origin

→ A curve represented by $y=f(x)$ is said to be symmetric about origin if it is odd function.

$$\text{i.e. } f(-x) = -f(x)$$

$$\text{i.e. } f(x) = x^3$$

$$f(-x) = (-x)^3$$

$$= -x^3$$

$$\text{or, } f(-x) = -f(x)$$

2) Test whether the following functions are symmetrical with respect to origin, co-ordinate axes or neither.

a) $y = x^2$

$$f(x) = x^2$$

$$f(-x) = (-x)^2$$

$$= x^2$$

This shows that $f(x)$ is even function hence it is symmetrical with respect to co-ordinate axes.

b) $y = x^5 + x^3 + x$

$$f(x) = x^5 + x^3 + x$$

$$f(-x) = (-x)^5 + (-x)^3 + (-x)$$

$$= -x^5 - x^3 - x$$

$$= -(x^5 + x^3 + x)$$

$$= -f(x)$$

This shows that $f(x)$ is an odd function hence it is symmetrical with respect to origin.

c) $y = \frac{x}{x^2 + 2}$

$$f(x) = \frac{x}{x^2 + 2}$$

$$f(-x) = \frac{-x}{(-x)^2 + 2} = \frac{-x}{x^2 + 2} = -\left(\frac{x}{x^2 + 2}\right)$$

$$f(-x) = -f(x)$$

This shows that $f(x)$ is an odd function, hence it is symmetrical with respect to origin.

d) $y = \frac{2}{x}$

$$f(x) = \frac{2}{x}$$

$$f(-x) = \frac{2}{-x} = -\left(\frac{2}{x}\right) = -f(x)$$

This shows that $f(x)$ is an odd function, hence it is symmetrical with respect to origin.

e) $y = 2 \sin 3x$

$$f(x) = 2 \sin 3x$$

$$\begin{aligned} f(-x) &= 2 \sin 3(-x) \\ &= -2 \sin 3x \\ &= -f(x) \end{aligned}$$

This shows that $f(x)$ is an odd function, hence it is symmetrical with respect to origin.

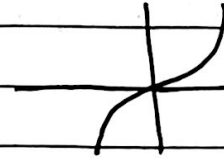
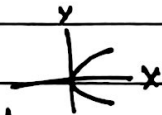
iii) Symmetric about x -axis

↳ A curve represented by $f(x, y) = 0$ is said to be symmetric about x -axis if $f(x, -y) = f(x, y)$

e.g.:

g) $y^2 = 4ax$

replace y by $-y$.



$$f(-y) = 4ax$$

Hence $f(x)$ is symmetric about x -axis.

h) $x^2 + y^2 = 25$

$$\Rightarrow y^2 = 25 - x^2$$

Replace y by $-y$

$$(-y)^2 = 25 - x^2$$

$$y^2 = 25 - x^2$$

Similarly,

$$x^2 = 25 - y^2$$

Replace x by $-x$.

$$(-x)^2 = 25 - y^2$$

$$x^2 = 25 - y^2$$

Hence $f(x)$ is symmetric about both x -axis and y -axis.

• Intercept

i) x -intercept, $y=0$

ii) y -intercept, $x=0$

3.

a) $y = 3x + 7$

For x -intercept $y=0$

$$3x + 7 = 0$$

$$3x = -7$$

$$x = -\frac{7}{3}$$

For y -intercept $x=0$

$$y = 0 + 7$$

$$\therefore y = 7$$

b) $y = x - 3$

For x -intercept $y=0$

$$x - 3 = 0$$

$$\therefore x = 3$$

For y -intercept $x=0$

$$y = 0 - 3$$

$$\therefore y = -3$$

c) $y = (x+1)(x-2)$

For x -intercept $y=0$

$$0 = (x+1)(x-2)$$

$$\Rightarrow x^2 - 2x + x - 2 = 0$$

$\Rightarrow$ Either,

$$x-2=0$$

$$x+1=0$$

$$x=2$$

$$x=-1$$

For y-intercept $x=0$

$$y=(0+1) \times (0-2)$$

$$y=1 \times -2$$

$$y=-2$$

4

d) $y = x^2 + 4x + 3$

For x-intercept, $y=0$

$$x^2 + 4x + 3 = 0$$

$$x^2 + 3x + x + 3 = 0$$

$$x(x+3) + 1(x+3) = 0$$

$$(x+3)(x+1) = 0$$

Either,

$$x+3=0$$

$$x+1=0$$

$$x=-3$$

$$x=-1$$

$$\therefore x = -1, -3$$

For y-intercept, $x=0$

$$y = 0 + 0 + 3$$

$$\therefore y = 3$$

e) $y = 16 - 4x^2$

For x-intercept, $y=0$

$$16 - 4x^2 = 0$$

$$4x^2 = 16$$

$$x^2 = 4$$

$$x = \sqrt{4}$$

$$\therefore x = \pm 2$$

For y-intercept, $x=0$

$$y = 16 - 0$$

$$\therefore y = 16$$

$$f, y = x(x-1)(x-2)$$

x-intercept, $y=0$

$$0 = x(x-1)(x-2)$$

Either,

$$x=0$$

$$x-1=0$$

$$x-2=0$$

$$x=1$$

$$x=2$$

$$\therefore x = 0, 1, 2$$

For y-intercept, $x=0$

$$y = 0(0-1)(0-2)$$

$$y = 0$$

$$\therefore y = 0$$

• Periodic Function

→ A function $f: A \rightarrow B$ is said to be periodic if $f(x+k) = f(x)$ for all $x \in A$

~~Here~~ Here smallest value of k is called period of the function.

Note: The trigonometric Functions, $\sin x$ and $\cos x$ are periodic with period 2π . but $\tan x$ is periodic with period π .

e.g: If $f(x) = \sin x$

$$\begin{aligned}\text{then } f(x+2\pi) &= \sin(x+2\pi) \\ &= \sin(2\pi+x) \\ &= \sin x\end{aligned}$$

$$f(x+2\pi) = f(x)$$

$\therefore f(x)$ is periodic with period 2π .

Q.

a) ~~$y = \sin 2x$~~

~~If $f(x) = \sin 2x$~~

$$\begin{aligned}\text{then } f(2x+2\pi) &= \sin 2(x+\pi) \\ &= \sin(2x+2\pi) \\ &= \sin 2x\end{aligned}$$

$$f(x+2\pi) = f(x)$$

$$\frac{x}{2} = \pi$$

$$x = 2\pi$$

~~$\therefore f(x)$ is periodic with period 2π~~

Q

b)

c) $y = 2 \tan \frac{x}{2}$

If $f(x) = 2 \tan \frac{x}{2}$

or, $f(x+\pi) = 2 \tan \left(\frac{x+\pi}{2} \right)$

or, $f(x+2\pi) = 2 \tan \left(\frac{x}{2} + \pi \right)$

or, $f(x+2\pi) = 2 \tan \frac{x}{2}$

$$f(x+2\pi) = f(x)$$

$\therefore f(x)$ is periodic with period 2π .

4.

a) $y = \sin 2x$

$$f(x) = \sin 2x$$

$$f(x + \pi) = \sin 2(x + \pi)$$

$$= \sin (2x + 2\pi)$$

$$= \sin 2x$$

$$= f(x)$$

$$\therefore f(x + \pi) = f(x)$$

Hence, $f(x)$ is periodic with period π .

~~b) $y = \cos(ax + b)$~~

~~$f(x) = \cos(ax + b)$~~

or ~~$f\left(\frac{x + 2\pi - b}{a}\right) = \cos\left(\frac{a\left(\frac{x + 2\pi - b}{a}\right) + b}{a}\right)$~~

~~$= \cos(2\pi - b + b)$~~

~~$= \cos(ax + 2\pi)$~~

$$\cos(ax + b) = y$$

$$ax = 2\pi$$

$$f(x) = \cos(ax + b)$$

$$x = \frac{2\pi}{a}$$

$$f\left(\frac{x + 2\pi}{a}\right) = \cos\left[\frac{a\left(\frac{x + 2\pi}{a}\right) + b}{a}\right]$$

$$= \cos(ax + 2\pi + b)$$

$$= \cos[2\pi + (ax + b)]$$

$$= \cos(ax + b)$$

$$\therefore f\left(\frac{x + 2\pi}{a}\right) = f(x)$$

Hence, $f(x)$ is periodic with period $\frac{2\pi}{a}$

• Increasing Function

→ A function $y = f(x)$ is said to be increasing if $x_2 > x_1 \Rightarrow f(x_2) > f(x_1)$ for x_1, x_2 belonging to domain of the function.

• Decreasing Function.

→ A function $y = f(x)$ is said to be decreasing if $x_2 > x_1 \Rightarrow f(x_2) < f(x_1)$ for x_1, x_2 belonging to domain of the function.

e.g.

5) a) $f(x) = 3 + 9x$

Let $x_1, x_2 \in \mathbb{R}$ such that $x_2 > x_1$.

$$\text{Then } x_2 > x_1 \Rightarrow 9x_2 > 9x_1$$

$$\Rightarrow 3 + 9x_2 > 3 + 9x_1$$

$$\text{i.e. } x_2 > x_1 \Rightarrow f(x_2) > f(x_1)$$

$\therefore f(x)$ is increasing function.

b) $f(x) = 7 - 3x$ is decreasing

Let $x_1, x_2 \in \mathbb{R}$ such that $x_2 > x_1$.

Then,

$$x_2 > x_1 \Rightarrow -3x_2 < -3x_1$$

$$\Rightarrow 7 - 3x_2 < 7 - 3x_1$$

$$\Rightarrow f(x_2) < f(x_1)$$

$\therefore f(x)$ is decreasing function.

6) Draw the graph of the following quadratic functions by indicating their characteristics.

a) $y = 3(x+2)^2$

i.e., $y = 3x^2 + 12x + 12$

Comparing with $y = ax^2 + bx + c$

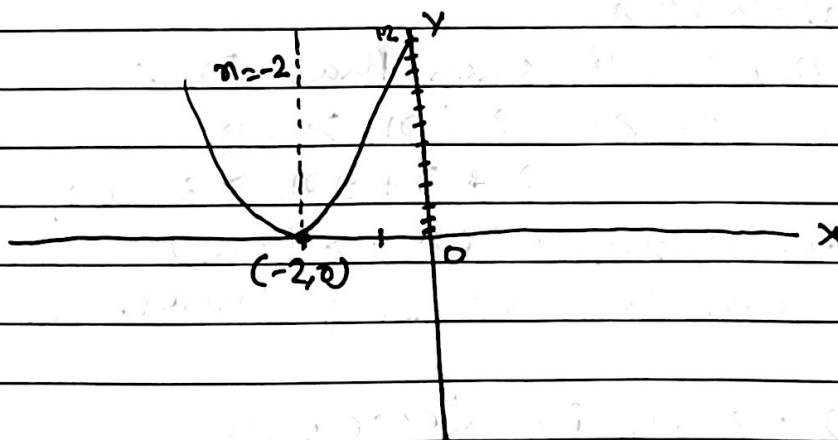
We have,

$$\text{Vertex} = \left(\frac{-b}{2a}, \frac{4ac - b^2}{4a} \right)$$

$$= \left(\frac{-12}{2 \times 3}, \frac{4 \times 3 \times 12 - 12^2}{4 \times 3} \right)$$

$$= \left(\frac{-12}{6}, \frac{144 - 144}{12} \right)$$

$\therefore \text{Vertex} = (-2, 0)$



For x -intercept, $y = 0$

$$3(x+2)^2 = 0$$

$$x+2 = 0$$

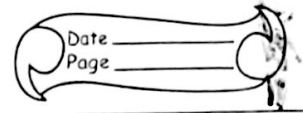
$$x = -2$$

For y -intercept, $x = 0$

$$y = 3(0+2)^2$$

$$= 3 \times 2^2$$

$a = +ve$ (upward)
 $a = -ve$ (downward)



$$\therefore y = 12$$

So curve cuts x -axis at $(-2, 0)$ and
 y -axis at $(0, 12)$

Since $A = 3 > 0$

So parabola turns upward.

For axis,

$$x = -\frac{b}{2a}$$

$$x = -2$$

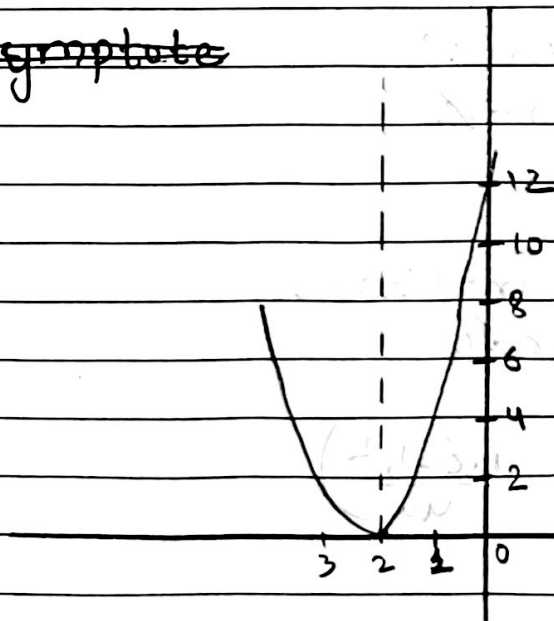
Domain = $(-\infty, \infty)$

Range = $[0, \infty)$

The graph for the given function
is given below:

Scale: In both axes, 10 small boxes = 1 unit

• ~~Asymptote~~



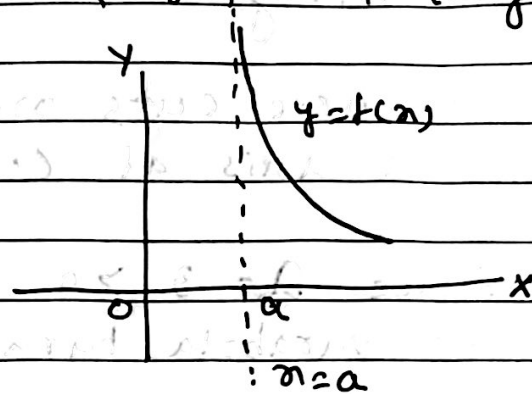


• Asymptote

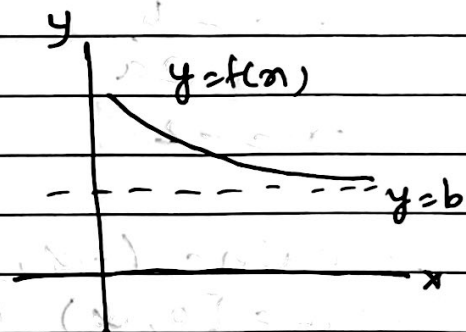
(Asymptote || to y-axis)

A line $x=a$ is said to be Asymptote to a curve, $y=f(x)$ if $y \rightarrow \infty$ where $x \rightarrow a$

(i.e. x close to a makes y close to ∞).



A line $y=b$ is said to be asymptote to a curve, $y=f(x)$ if $x \rightarrow \infty$ when $y \rightarrow b$.



i.e. (y close to b makes x close to ∞). (Asymptote || to x-axis)

6.

$$b) y = 12 - 3(x-2)^2$$

$$\text{i.e. } y = 12 - 3(x^2 - 4x + 4)$$

$$= 12 - 3x^2 + 12x - 12$$

$$= 12x - 3x^2$$

$$= -3x^2 + 12x$$

Comparing with $ax^2 + bx + c$

$$a = -3, b = 12, c = 0$$

We have,

$$\text{Vertex} = \left(\frac{-b}{2a}, \frac{4ac - b^2}{4a} \right)$$

$$\begin{aligned} \text{Vertex} &= \left(\frac{-12}{2 \times (-3)}, \frac{4 \times (-3) \times 0 - 12^2}{4 \times (-3)} \right) \\ &= \left(\frac{-12}{-6}, \frac{0 - 144}{-12} \right) \\ &= (2, 12) \end{aligned}$$

$$\therefore \text{Vertex} = (2, 12)$$

For x -intercept, $y = 0$

$$0 = 12 - 3(x-2)^2$$

$$\text{or, } 0 = -3x^2 + 12x$$

$$\text{or, } 3x^2 = 12x$$

$$\text{or, } 3x^2 - 12x = 0$$

$$\text{or, } 3x(x-4) = 0$$

Either,

$$x = 0, 4$$

For y -intercept, $x = 0$

$$y = -3 \times 0^2 + 12 \times 0$$

$$y = 0$$

So curve cuts x -axis at $(0, 0)$ and $(4, 0)$
and y -axis at $(0, 0)$.

Since $a < 0$ [$-3 < 0$]

So the parabola turns downward

For axis

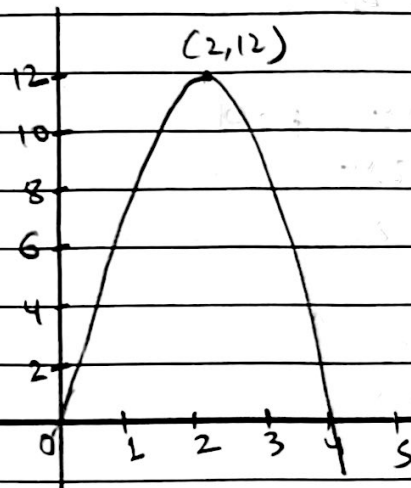
$$x = -\frac{b}{2a}$$

$$n = 0, 4$$

$$\text{Domain} = (-\infty, \infty)$$

$$\text{Range} = (-\infty, 12]$$

The graph for the given function is



$$c) y = 2x^2 - 8x + 6$$

Comparing with $ax^2 + bx + c$ we get,
 $a = 2, b = -8, c = 6$

We have,

$$\text{Vertex} = \left(\frac{-b}{2a}, \frac{4ac - b^2}{4a} \right)$$

$$= \left(\frac{-(-8)}{2 \times 2}, \frac{4 \times 2 \times 6 - (-8)^2}{4 \times 2} \right)$$

$$= \left(\frac{+8}{4}, \frac{48 - 64}{8} \right)$$

$$= (2, -2)$$

For x -intercept, $y = 0$

$$0 = 2x^2 - 8x + 6$$

$$\Rightarrow x^2 - 4x + 3 = 0$$

$$\Rightarrow x^2 - 3x - x + 3 = 0$$

$$\Rightarrow x(x-3) - 1(x-3) = 0$$

$$\Rightarrow (x-1)(x-3) = 0$$

Either,

$$x-1=0$$

$$x-3=0$$

$$x=1$$

$$x=3$$

$$\therefore x=1, 3$$

For y-intercept, $x=0$

$$y = 2 \times 0^2 - 8 \times 0 + 6$$

$$\therefore y=6$$

So curve cuts x-axis at $(1,0)$ and $(3,0)$
and y-axis at $(0,6)$

Since $a=2 > 0$

So the parabola turns upward.

For axis,

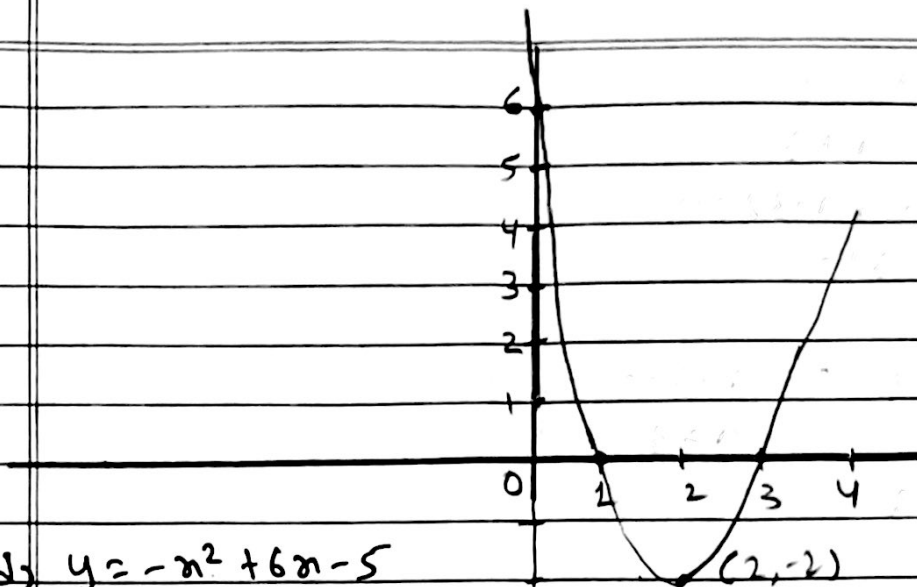
$$x = -\frac{b}{2a}$$

$$x = 1, 3$$

$$\text{Domain} = (-\infty, \infty)$$

$$\text{Range} = [-2, \infty)$$

The graph for the given function is



d) $y = -x^2 + 6x - 5$

Comparing with $ax^2 + bx + c$

$a = -1, b = 6, c = -5$

We have,

$$\text{Vertex} = \left(\frac{-b}{2a}, \frac{4ac - b^2}{4a} \right)$$

$$= \left(\frac{-6}{2 \times -1}, \frac{4 \times (-1) \times (-5) - 6^2}{4 \times (-1)} \right)$$

$$= \left(\frac{-6}{-2}, \frac{20 - 36}{-4} \right)$$

$$= (3, 4)$$

For x-intercept $y = 0$.

$$0 = -x^2 + 6x - 5$$

$$\Rightarrow x^2 - 6x + 5 = 0$$

$$\Rightarrow x^2 - 5x - x + 5 = 0$$

$$\Rightarrow x(x-5) - 1(x-5) = 0$$

$$\Rightarrow (x-1)(x-5) = 0$$

Either,

$$x-1=0$$

$$x-5=0$$

$$x=1$$

$$x=5$$

$$\therefore x = 1, 5$$

For y-intercept, $x=0$

$$y = -0^2 + 6 \times 0 - 5$$

$$y = -5$$

So curve cuts x-axis at $(1,0)$ and $(5,0)$ and y-axis at $(0,-5)$

Since $a = -1 < 0$

so the parabola turns downward

For axis,

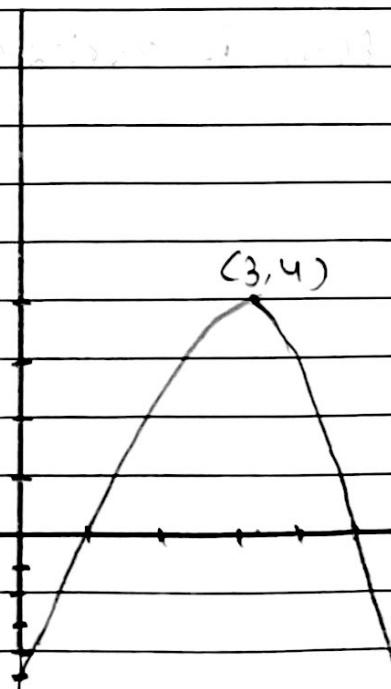
$$x = \frac{-b}{2a}$$

$$x = 1.5$$

Domain = $(-\infty, \infty)$

Range = $(-\infty, 4]$

The graph for the given function is



4.

b) $y = \cos nx$

$nx = 2\pi$

$f(x) = \cos nx$

$n = 2$

$f(x+2) = \cos n(x+2)$

$= \cos (nx + 2n)$

$= \cos (2\pi + nx)$

$= \cos nx$

$= f(x)$

$\therefore f(x+2) = f(x)$

Hence, $f(x)$ is periodic with period 2.

c) $y = \sin x + \cos x$

let $f(x) = \sin x + \cos x$

$f(x+2\pi) = \sin (x+2\pi) + \cos (x+2\pi)$

$= \sin (2\pi + x) + \cos (2\pi + x)$

$= \sin x + \cos x$

$= f(x)$

$\therefore f(x+2\pi) = f(x)$

Hence, $f(x)$ is periodic with period 2π .